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CASCODE SEMICONDUCTOR DEVICE

Abstract

A semiconductor power device may include a first power transistor configured to generate an output voltage. A semiconductor power device may include a second power transistor configured to receive an input voltage. The second power transistor is connected to the first power transistor in a cascode configuration. A semiconductor power device may include a voltage divider circuit connected to a gate terminal of the first power transistor and a source terminal of the second power transistor.

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Background/Summary

BACKGROUND

[0001] The on-resistance of a typical unipolar semiconductor power device may scale

proportionally to the square of the rated voltage of the semiconductor power device. The on-resistance may be a measure of the resistance of the transistor when the transistor is conducting current, while the breakdown voltage is a limit (e.g., a maximum voltage) that the transistor can withstand without malfunctioning (e.g., thermal runaway or destruction). In general, there is a direct relationship between the breakdown voltage and the on-resistance (e.g., as the breakdown voltage of the transistor increases, on-resistance of a transistor may increase).

SUMMARY

[0002] In some aspects, the techniques described herein relate to a semiconductor power device including: a first power transistor configured to generate an output voltage; a second power transistor configured to receive an input voltage, the second power transistor connected to the first power transistor in a cascode configuration; and a voltage divider circuit connected to a gate terminal of the first power transistor and a source terminal of the second power transistor.

[0003] In some aspects, the techniques described herein relate to a semiconductor power device including: a first power transistor configured to generate an output voltage; a second power transistor configured to receive an input voltage, the second power transistor connected to the first power transistor in a cascode configuration; and a voltage divider circuit configured to indirectly couple a gate terminal of the first power transistor to a source terminal of the second power transistor, the voltage divider circuit including at least one first circuit element and at least one second circuit element.

[0004] In some aspects, the techniques described herein relate to a method including: activating a first power transistor of a semiconductor power device and a second power transistor of the semiconductor power device in response to a gate-source voltage of the semiconductor power device being equal to or greater than a threshold voltage of the second power transistor; deactivating the second power transistor in response to the gate-source voltage of the semiconductor power device being less than the threshold voltage of the second power transistor; and reducing, by a voltage divider circuit, a potential of a gate-source voltage of the first power transistor to be less than a gate-source maximum voltage rating, the voltage divider circuit indirectly connecting a gate of the first power transistor to a source terminal of the second power transistor.

[0005] The details of one or more implementations are set forth in the accompanying drawings and the description below. Other features will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1A illustrates a semiconductor power device with a voltage divider and first and second power transistors connected in a cascode configuration according to an aspect.

[0007] FIG. 1B illustrates a graph depicting a reduction in specific on-resistance for a high voltage semiconductor power device according to an aspect.

[0008] FIG. 2A illustrates a semiconductor power device with a voltage divider and first and second power transistors connected in a cascode configuration according to another aspect.

[0009] FIGS. 2B and 2C illustrate voltage and current waveforms of a semiconductor power device and first and second power transistors according to an aspect.

[0010] FIG. 3A illustrates a semiconductor power device with a voltage divider and first and second power transistors connected in a cascode configuration according to another aspect.

[0011] FIGS. 3B and 3C illustrate voltage and current waveforms of a semiconductor power device and first and second power transistors according to an aspect.

[0012] FIG. 4A illustrates a semiconductor power device with a voltage divider and first and

second power transistors connected in a cascode configuration according to another aspect.

[0013] FIGS. 4B and 4C illustrate voltage waveforms of a semiconductor power device and first and second power transistors according to an aspect.

[0014] FIG. 5 illustrates a flowchart depicting example operations of a semiconductor power device according to an aspect.

DETAILED DESCRIPTION

[0015] The present disclosure relates to a semiconductor power device with a first power transistor (e.g., an upper transistor) and a second power transistor (e.g., a lower transistor), where a terminal (e.g., a gate) of the first power transistor is coupled (e.g., indirectly coupled) to a terminal (e.g., a source) of the second power transistor via a voltage divider. The semiconductor power device discussed herein may have low on-resistance while providing a relatively high breakdown voltage (e.g., improving the tradeoff between on-resistance and breakdown voltage), thereby providing a more efficient, compact, and/or reliable power electronics system. In some examples, the semiconductor power device discussed herein may provide shorter termination lengths, smaller active area and thinner epitaxial layers, which can contribute to higher yield and/or lower manufacturing cost.

[0016] The first power transistor and the second power transistor are connected to each other in a cascode configuration (e.g., the drain terminal of the second power transistor is connected to the source of the first power transistor, while the gate of the first transistor is coupled to the source of the second transistor). The voltage divider includes at least one first circuit element (e.g., one or more first circuit elements) and at least one second circuit element (e.g., one or more second circuit element). In some examples, the first circuit element is a voltage clamping element. In some examples, the first circuit element includes one or more diodes. In some examples, the second circuit element is a capacitor. In some examples, the second circuit element includes a capacitor and a resistor. According to the techniques discussed herein, the gate-source voltage (V_{gs_upper}) of the first power transistor may be less (e.g., always less) than a maximum allowed gate-source voltage ($V_{gs_upper_max}$). In some examples, the second power transistor may be a high voltage device. As a result, the drain-source voltage (V_{ds_lower}) of the second power transistor may substantially exceed the gate-source voltage (V_{gs_upper}) of the first power transistor. In some examples, the voltage rating of the semiconductor power device is the sum of the voltage ratings of the first and second power transistors.

[0017] In some conventional approaches, the lower device of the cascode may be a low voltage transistor, which may allow the gate of an upper device to be directly connected to the source of a lower device (e.g., V_{ds_lower} is always $< V_{gs_max_upper}$). In contrast to some conventional approaches, the semiconductor power device discussed herein may include a high voltage lower transistor, thus V_{ds_lower} may be higher (e.g., substantially higher) than $V_{gs_max_upper}$. In some examples, the gate of the upper transistor directly connected to the source of the lower device may lead to the upper transistor damage and/or whole device damage. However, as discussed herein, the gate of the first power transistor is coupled (e.g. indirectly connected) to the source of the second power transistor using a voltage divider, where one of the elements of the voltage divider includes a voltage clamping element, which can reduce V_{gs} applied to the upper transistor to be less than a $V_{gs_max_upper}$ (e.g., a gate-source voltage maximum rating), while still enabling a cascode operation.

[0018] In some examples, the first power transistor includes a junction field-effect transistor (JFET), a metal-oxide-semiconductor field-effect transistor (MOSFET, or a bipolar junction transition (BJT). In some examples, the first power transistor includes high-voltage silicon carbide JFET. In some examples, the first power transistor is configured as a normally-on device. A normally-on device is a transistor that conducts current when the gate-to-source voltage is zero or substantially zero. The first power transistor's gate-to-source voltage can be used to control the amount of current that the device conducts. In some examples, the second power transistor includes

MOSFET, a JFET, or a BJT. In some examples, the second power transistor includes a high-voltage silicon carbide MOSFET. In some examples, the second power transistor is configured as a normally-off device. A normally-off device is a transistor that does not conduct current when the gate-to-source voltage is zero or substantially zero. In some examples, the first power transistor is a normally-off device and the second power transistor is a normally-on device.

[0019] FIGS. 1A to 1B illustrate a semiconductor power device **100** according to an aspect. The semiconductor power device **100** includes a first power transistor **108** and a second power transistor **110**. The second power transistor **110** is configured to receive an input voltage. The first power transistor **108** is configured to generate an output voltage. The first power transistor **108** and the second power transistor **110** are connected in a cascode configuration. A cascode configuration may be a two stage amplifier circuitry in which two or more semiconductor power devices (e.g., the first power transistor **108** and the second power transistor **110**) are connected in series to achieve higher input-output isolation, higher input impedance, high output impedance, and/or higher bandwidth. The semiconductor power device **100** discussed herein may also allow a higher voltage and/or current rating. In some examples, the semiconductor power device **100** is referred to as a cascode device. In some examples, the semiconductor power device **100** is referred to as a high voltage cascode device.

[0020] When a voltage is applied across the series combination, the voltage is divided between the first power transistor **108** and the second power transistor **110**, which may allow the cascode device to handle higher voltages than each transistor could support individually. Similarly, when a cascode device is designed for a specific voltage rating, the series combination of two high voltage devices may allow a lower total resistance (e.g. the sum of on-state resistances of two transistors) as compared to the resistance of a single transistor having the same/similar active area (e.g., if designed for the same rated voltage). This may allow the high voltage cascode device to handle higher currents as compared to a single unipolar power switch. In some examples, the semiconductor power device **100** includes a cascode connection of two or more semiconductor power devices to construct a high-voltage device having a breakdown voltage of $BV = BV_{\text{sub.Device1}} + BV_{\text{sub.Device2}}$ and an on-resistance of $R_{\text{on}} = R_{\text{on.sub.Device1}} + R_{\text{on.sub.Device2}}$.

[0021] The first power transistor **108** may be referred to as an upper device because, in some examples, the first power transistor **108** is located closer to the output voltage source. The second power transistor **110** may be referred to as a lower device because, in some examples, the second power transistor **110** is located closer to the input. In some examples, the first power transistor **108** is a higher voltage device than the second power transistor **110**. In some examples, the first power transistor **108** has a breakdown voltage that is higher than the breakdown voltage of the second power transistor **110**.

[0022] In some examples, the first power transistor **108** includes a junction field-effect transistor (JFET). In some examples, the first power transistor **108** includes a metal-oxide-semiconductor field-effect transistor (MOSFET). In some examples, the first power transistor **108** includes a bipolar junction transistor (BJT). In some examples, the first power transistor **108** includes a high-voltage (HV) silicon carbide (SiC) JFET. In some examples, the first power transistor is configured as a normally-on device, where the first power transistor **108** conducts current when the gate-to-source voltage is zero or substantially zero. In some examples, the second power transistor **110** includes a MOSFET. In some examples, the second power transistor **110** includes a HV SiC MOSFET. In some examples, the second power transistor **110** includes a JFET. In some examples, the second power transistor **110** includes a BJT. In some examples, the second power transistor **110** is configured as a normally-off device, where the second power transistor **110** does not conduct current when the gate-to-source voltage is zero or substantially zero. In some examples, the first power transistor **108** and the second power transistor **110** are power transistors of different types (one is a MOSFET, and the other is a JFET or BJT). In some examples, the first power transistor

108 and the second power transistor **110** are transistors of the same type. In some examples, a transistor (e.g., the first power transistor **108** or the second power transistor **110**) is a BJT with terminals referred to as a base, an emitter and a collector. As discussed herein, the terms gate terminal, source terminal, and drain terminal may encompass and/or be replaced with base (or base terminal), emitter (or emitter terminal), and collector (or collector terminal), respectively.

[0023] The first power transistor **108** includes a drain terminal **101**, a source terminal **103**, and a gate terminal **105**. The second power transistor **110** includes a drain terminal **107**, a source terminal **109**, and a gate terminal **111**. In some examples, the second power transistor **110** includes a diode **116** (e.g. internal body diode) connected to the drain terminal **107** of the second power transistor **110** and the source terminal **109** of the second power transistor **110**. In some examples, the drain terminal **101** of the first power transistor **108** serves as a drain terminal of the cascode device. The source terminal **103** of the first power transistor **108** is connected to the drain terminal **107** of the second power transistor **110**. In some examples, the source terminal **109** of the second power transistor **110** serves as a source terminal of the cascode device. In some examples, the gate terminal **111** of the second power transistor **110** serves as a gate terminal of the cascode device.

[0024] The gate terminal **105** of the first power transistor **108** is coupled (e.g., indirectly coupled) to the source terminal **109** of the second power transistor **110** via a voltage divider circuit **120**. In other words, the gate terminal **105** of the first power transistor **108** is not directly connected to the source terminal **109** of the second power transistor **110**. Rather, the voltage divider circuit **120** indirectly connects the gate terminal **105** of the first power transistor **108** to the source terminal **109** of the second power transistor **110**. The gate terminal **105** of the first power transistor **108** is connected to the voltage divider circuit **120**. The source terminal **109** of the second power transistor **110** is connected to the voltage divider circuit **120**.

[0025] The voltage divider circuit **120** includes a first circuit element **112** (D1) and a second circuit element **114** (D2). The first circuit element **112** may connect the source terminal **103** of the first power transistor **108** and the gate terminal **105** of the first power transistor **108**. A first terminal **113** of the first circuit element **112** is connected to the source terminal **103** of the first power transistor **108**. The first terminal **113** of the first circuit element **112** is connected to the drain terminal **107** of the second power transistor **110**. A second terminal **115** of the first circuit element **112** is connected to the gate terminal **105** of the first power transistor **108**. The second terminal **115** of the first circuit element **112** is connected to a first terminal **117** of the second circuit element **114**. The first terminal **117** of the second circuit element **114** is connected to the gate terminal **105** of the first power transistor **108**. A second terminal **119** of the second circuit element **114** is connected to the source terminal **109** of the second power transistor **110**.

[0026] In some examples, the first circuit element **112** is a voltage clamping (e.g. voltage limiting) element. In some examples, the first circuit element **112** is a diode. In some examples, the first circuit element **112** is a Zenner diode. In some examples, the first circuit element **112** is a p-n diode. In some examples, the first circuit element **112** is a Schottky diode. In some examples, the first circuit element **112** includes two or more diodes. In some examples, the first circuit element **112** includes two or more diodes connected in series. In some examples, the first circuit element **112** is configured to cause the gate-source voltage (V_{gs_upper}) of the first power transistor **108** to be less (e.g., always less) than a maximum gate-source voltage ($V_{gs_upper_max}$) (e.g., a gate-source breakdown voltage).

[0027] In some examples, the second circuit element **114** is a capacitor. In some examples, the second circuit element **114** includes two or more capacitors. In some examples, the second circuit element **114** includes a capacitor and a resistor. In some examples, the resistor and the capacitor are connected in parallel. In some examples, the second circuit element **114** includes one or more capacitors and/or one or more resistors. In some examples, the gate-source voltage (V_{gs_upper}) of the first power transistor **108** is independent of the drain-source voltage (V_{ds_lower}) of the second power transistor **110**. As a result, the second power transistor **110**

may be a high voltage device. In some examples, the voltage rating of the semiconductor power device **100** is the sum of the voltage rating of the first power transistor **108** and the voltage rating of the second power transistor **110**.

[0028] In some examples, the first circuit element **112** is integrated into the upper device. For example, the semiconductor power device **100** includes a semiconductor package **102** with a first semiconductor device **104** and a second semiconductor device **106**. In some examples, the second semiconductor device **106** is a device that is separate and/or distinct from the first semiconductor device **104**. The second semiconductor device **106** may be connected to the first semiconductor device **104**. In some examples, the first semiconductor device **104** is referred to as an upper device. In some examples, the first semiconductor device **104** is a semiconductor die (e.g., a portion of semiconductor material wafer) that includes the first power transistor **108**. In some examples, the first semiconductor device **104** includes the first circuit element **112**. In some examples, the second circuit element **114** is integrated into the lower device. In some examples, the second semiconductor device **106** is referred to as a lower device. In some examples, the second semiconductor device **106** is a semiconductor die (e.g., a silicon wafer) that includes the second power transistor **110**. In some examples, the second semiconductor device **106** includes the second circuit element **114**.

[0029] In response to the gate-source voltage ($V_{\text{sub.gs_cascode}}$) of the semiconductor power device **100** being greater than a threshold voltage ($V_{\text{sub.th_second_transistor}}$) of the second power transistor **110**, the first power transistor **108** and the second power transistor **110** are activated (e.g., turned-on), thereby conducting a current ($I_{\text{sub.d_cascode}}$) through both the first power transistor **108** and the second power transistor **110**, where

$$I_{\text{sub.d_cascode}} = V_{\text{sub.ds_cascode}} / (R_{\text{sub.ds(on)_second_transistor}} + R_{\text{sub.ds(on)_first_transistor}})$$

 $V_{\text{sub.ds_cascode}}$ is the drain-source voltage applied to the cascode device.

$R_{\text{sub.ds(on)_second_transistor}}$ is the on-resistance of the second power transistor **110**.

$R_{\text{sub.ds(on)_first_transistor}}$ is the on-resistance of the first power transistor **108**.

[0030] In response to the gate-source voltage ($V_{\text{sub.gs_cascode}}$) of the semiconductor power device **100** being less than the voltage threshold ($V_{\text{sub.th_second_transistor}}$) of the second power transistor **110**, the second power transistor **110** is deactivated (e.g., turned off), thereby increasing the potential of the drain terminal **107** of the second power transistor **110** and increasing the potential of the source terminal **103** of the first power transistor **108**. The increasing potential of the drain terminal **107** and the increasing potential of the source terminal **103** may cause the gate-source voltage of the first power transistor **108** to become negative, effectively deactivating the first power transistor **108**, which would share applied voltage as

$$V_{\text{sub.ds_cascode}} = V_{\text{sub.ds_first_transistor}} + V_{\text{sub.ds_second_transistor}}$$

 $V_{\text{sub.ds_first_transistor}}$ is the drain-source voltage across the first power transistor **108**. $V_{\text{sub.ds_second_transistor}}$ is the drain-source voltage across the second power transistor **110**.

[0031] In some examples, the maximum allowed gate-source voltage ($V_{\text{sub.gs_max_first_transistor}}$) of the first power transistor **108** is less than (e.g., substantially less than) than the drain-source voltage ($V_{\text{sub.ds_second_transistor}}$) of the second power transistor **110** in the off-state. In some examples, since the maximum allowed gate-source voltage ($V_{\text{sub.gs_max_first_transistor}}$) is less than (e.g., substantially less than) than the drain-source voltage ($V_{\text{sub.ds_second_transistor}}$) in the off-state, the gate-source voltage ($V_{\text{sub.gs_first_transistor}}$) of the first power transistor **108** is limited by the first circuit element **112** (D1) of the voltage divider circuit **120** to avoid (e.g., prevent) device malfunction (e.g., destruction). The remaining voltage may drop over the second circuit element **114** (D2) of the voltage divider (e.g., $D2: V_{\text{sub.D2}} = V_{\text{sub.ds_second_transistor}} - V_{\text{sub.D1}}$).

[0032] FIG. 1B illustrates a graph that depicts the specific on-resistance as a function of a breakdown voltage of the unipolar semiconductor (e.g., SiC) power device **100** of FIG. 1A and several single transistor devices according to an aspect. Line **136** depicts an example of a unipolar

limit (e.g., SiC unipolar limit). Line **132** depicts an example of the specific on-resistance as a function of the breakdown voltage for the semiconductor power device **100** of FIG. **1A**. Line **134** depicts an example of a specific on-resistance as a function of the breakdown voltage for a single JFET device (e.g., SiC JET). Line **130** depicts an example of a specific on-resistance as a function of the breakdown voltage for a single MOSFET device (e.g., SiC MOSFET). As shown in FIG. **1B**, the semiconductor power device **100** of FIG. **1A** may achieve a reduction in the amount of on-resistance as compared to single SiC devices having the same maximum voltage rating (e.g. breakdown voltage).

[0033] FIG. **2A** illustrates a semiconductor power device **200** according to an aspect. The semiconductor power device **200** may be an example of the semiconductor power device **100** of FIGS. **1A** and **1B** and may include any of the details discussed herein. The semiconductor power device **200** includes a first power transistor **208** and a second power transistor **210**. The first power transistor **208** and the second power transistor **210** are connected in a cascode configuration. A cascode configuration may be a circuit topology in which two or more semiconductor power devices (e.g., the first power transistor **208** and the second power transistor **210**) are connected in series to achieve a higher voltage and/or current rating. When a voltage is applied across the series combination, the voltage is divided across each of the first power transistor **208** and the second power transistor **210**, which may allow the devices to handle higher voltages than they could handle individually. Similarly, as indicated in FIG. **1B**, when cascode device is designed for a specific voltage rating, the series combination of two high voltage devices allows lower total resistance (e.g. the sum of on-state resistances of two transistors) compared to the resistance of the single transistor having same/similar active area, if designed for the same rated voltage. This allows the high voltage cascode device to handle higher currents compared to single unipolar power switches. In some examples, the semiconductor power device **200** includes a cascode connection of two or more semiconductor power devices to construct a high-voltage device having a breakdown voltage of $BV=BV_{\text{sub.Device1}}+BV_{\text{sub.Device2}}$ and an on-resistance of $R_{\text{on}}=R_{\text{on.sub.Device1}}+R_{\text{on.sub.Device2}}$.

[0034] As shown in FIG. **2A**, the gate terminal of the first power transistor **208** is coupled (e.g., indirectly coupled) to the source terminal of the second power transistor **210** via a voltage divider circuit **220**. In other words, the gate terminal of the first power transistor **208** is not directly connected to the source terminal of the second power transistor **210**. Rather, the voltage divider circuit **220** indirectly connects the gate terminal of the first power transistor **208** to the source terminal of the second power transistor **210**. The gate terminal of the first power transistor **208** is connected to the voltage divider circuit **220**. The source terminal of the second power transistor **210** is connected to the voltage divider circuit **220**.

[0035] As shown in FIG. **2A**, the voltage divider circuit **220** includes a diode **212** and a capacitor **214**. The diode **212** may be an example of the first circuit element **112** of FIG. **1A**. The capacitor **214** may be an example of the second circuit element **114** of FIG. **1A**. The diode **212** may be connected to the source terminal of the first power transistor **208** and the gate terminal of the first power transistor **208**. A first terminal of the diode **212** is connected to the source terminal of the first power transistor **208**. The first terminal of the diode **212** (e.g. cathode) is connected to the drain terminal of the second power transistor **210**. A second terminal of the diode **212** (e.g. anode) is connected to the gate terminal of the first power transistor **208**. The second terminal of the diode **212** is connected to a first terminal of the capacitor **214**. The first terminal of the capacitor **214** is connected to the gate terminal of the first power transistor **208**. A second terminal of the capacitor **214** is connected to the source terminal of the second power transistor **210**.

[0036] In some examples, the diode **212** is a voltage clamping element. In some examples, the diode **212** is a Zenner diode. In some examples, the diode **212** is a Zenner diode having a breakdown voltage that is less than the maximum gate-source voltage (e.g. gate-source breakdown voltage) of the first power transistor **208**. In some examples, the gate-source voltage of the first

power transistor **208** is clamped (e.g. limited) by the breakdown voltage of the diode **212**.

[0037] In some examples, the diode **212** is integrated into the upper device. For example, the semiconductor power device **200** includes a semiconductor package **202** with a first semiconductor device **204** and a second semiconductor device **206**. In some examples, the second semiconductor device **206** is a device that is separate and/or distinct from the first semiconductor device **204**. The second semiconductor device **206** may be connected to the first semiconductor device **204**. In some examples, the first semiconductor device **204** is referred to as an upper device. In some examples, the first semiconductor device **204** is a semiconductor die (e.g., a portion of semiconductor material wafer) that includes the first power transistor **208**. In some examples, the first semiconductor device **204** includes the diode **212**. In some examples, the capacitor **214** is integrated into the lower device. In some examples, the second semiconductor device **206** is referred to as a lower device. In some examples, the second semiconductor device **206** is a semiconductor die (e.g., a silicon wafer) that includes the second power transistor **210**. In some examples, the second semiconductor device **206** includes the capacitor **214**. In some embodiments the diode **212** and capacitor **214** are integrated into one of the power devices (e.g. into the first power transistor **208** or the second power transistor **210**). In some embodiments the diodes **312** and capacitor **314** can be standalone elements interconnected with power transistors **308** and **310** inside the package.

[0038] FIGS. 2B and 2C illustrate example waveforms of the semiconductor power device **200**, including the first power transistor **208** and the second power transistor **210**. As shown in FIGS. 2B and 2C, the first power transistor **208** and the second power transistor **210** may operate below their maximum rated voltages efficiently sharing total cascode device voltage between each other.

[0039] FIG. 3A illustrates a semiconductor power device **300** according to an aspect. The semiconductor power device **300** may be an example of the semiconductor power device **100** of FIGS. 1A and 1B and may include any of the details discussed herein. The semiconductor power device **300** includes a first power transistor **308** and a second power transistor **310**. The first power transistor **308** and the second power transistor **310** are connected in a cascode configuration. A cascode configuration may be a circuit topology in which two or more power semiconductor devices (e.g., the first power transistor **308** and the second power transistor **310**) are connected in series to achieve a higher voltage and/or current rating. When a voltage is applied across the series combination, the voltage is divided across each of the first power transistor **308** and the second power transistor **310**, which may allow the devices to handle higher voltages than they could individually. Similarly, when cascode device is designed for a specific voltage rating, the series combination of two high voltage devices allows lower total resistance (e.g. the sum of on-state resistances of two transistors) compared to the resistance of the single transistor having same/similar active area, if designed for the same rated voltage. This allows the high voltage cascode device to handle higher currents compared to single unipolar power switches. In some examples, the semiconductor power device **300** includes a cascode connection of two or more semiconductor devices to construct a high-voltage device having a breakdown voltage of $BV=BV_{\text{sub.Device1}}+BV_{\text{sub.Device2}}$ and an on-resistance of $R_{\text{on}}=R_{\text{on.sub.Device1}}+R_{\text{on.sub.Device2}}$.

[0040] As shown in FIG. 3A, the gate terminal of the first power transistor **308** is coupled (e.g., indirectly coupled) to the source terminal of the second power transistor **310** via a voltage divider circuit **320**. In other words, the gate terminal of the first power transistor **308** is not directly connected to the source terminal of the second power transistor **310**. Rather, the voltage divider circuit **320** indirectly connects the gate terminal of the first power transistor **308** to the source terminal of the second power transistor **310**. The gate terminal of the first power transistor **308** is connected to the voltage divider circuit **320**. The source terminal of the second power transistor **310** is connected to the voltage divider circuit **320**.

[0041] As shown in FIG. 3A, the voltage divider circuit **320** includes a plurality of diodes **312** (e.g., a sequence or a chain of diodes **312**) and a capacitor **314**. The diodes **312** may be an example of the

first circuit element **112** of FIG. **1A**. In some examples, the voltage divider circuit **320** includes two or more diodes **312** connected in series with each other. In some examples, the diodes **312** includes p-n diodes. In some examples, the diodes **312** include Schottky diodes. The diodes **312** may include a diode **312-1**, a diode **312-2**, a diode **312-3**, and a diode **312-4**. However, it is noted that the number of diodes **312** may be two, three, or any number greater than four. In some examples, the diodes **312** may be configured such as

$$[00001] n \times V_f < V_{gs_max_transistor_first},$$

where n is the number of diodes in the chain and V_f is the forward voltage drop (e.g. turn-on voltage) of a single diode. In some examples, the clamping voltage is defined by the sum of the turn-on voltages of the diodes **312**.

[0042] In some examples, the diodes **312** are integrated into the upper device. For example, the semiconductor power device **300** includes a semiconductor package **302** with a first semiconductor device **304** and a second semiconductor device **306**. In some examples, the second semiconductor device **306** is a device that is separate and/or distinct from the first semiconductor device **304**. The second semiconductor device **306** may be connected to the first semiconductor device **304**. In some examples, the first semiconductor device **304** is referred to as an upper device. In some examples, the first semiconductor device **304** is a semiconductor die (e.g., a silicon wafer) that includes the first power transistor **308**. In some examples, the first semiconductor device **304** includes the diodes **312**. In some examples, the capacitor **314** is integrated into the lower device. In some examples, the second semiconductor device **306** is referred to as a lower device. In some examples, the second semiconductor device **306** is a semiconductor die (e.g., a portion of semiconductor material wafer) that includes the second power transistor **310**. In some examples, the second semiconductor device **306** includes the capacitor **314**. In some examples, the diodes **312** and capacitor **314** may be integrated into one of the power devices (e.g. into the first power transistor **308** or the second power transistor **310**). In some embodiments the diodes **312** and capacitor **314** can be standalone elements interconnected with power transistors **308** and **310** inside the package.

[0043] FIGS. **3B** and **3C** illustrate example waveforms of the semiconductor power device **300**, including the first power transistor **308** and the second power transistor **310**. As shown in FIGS. **3B** and **3C**, the first power transistor **308** and the second power transistor **310** may operate below their maximum rated voltages efficiently sharing total cascode device voltage between each other.

[0044] FIG. **4A** illustrates a semiconductor power device **400** according to an aspect. The semiconductor power device **400** may be an example of the semiconductor power device **100** of FIGS. **1A** and **1B** and may include any of the details discussed herein. The semiconductor power device **400** includes a first power transistor **408** and a second power transistor **410**. The first power transistor **408** and the second power transistor **410** are connected in a cascode configuration. A cascode configuration may be a circuit topology in which two or more power semiconductor devices (e.g., the first power transistor **408** and the second power transistor **410**) are connected in series to achieve a higher voltage and/or current rating. When a voltage is applied across the series combination, the voltage is divided across each of the first power transistor **408** and the second power transistor **410**, which may allow the devices to handle higher voltages than they could individually. Similarly, when cascode device is designed for a specific voltage rating, the series combination of two high voltage devices allows lower total resistance (e.g. the sum of on-state resistances of two transistors) compared to the resistance of the single transistor having same/similar active area, if designed for the same rated voltage. This allows the high voltage cascode device to handle higher currents compared to single unipolar power switches. In some examples, the semiconductor power device **400** includes a cascode connection of two or more semiconductor devices to construct a high-voltage device having a breakdown voltage of $BV=BV_{sub.Device1}+BV_{sub.Device2}$ and an on-resistance of $R_{on}=R_{on.sub.Device1}+R_{on.sub.Device2}$.

[0045] As shown in FIG. 4A, the gate terminal of the first power transistor **408** is coupled (e.g., indirectly coupled) to the source terminal of the second power transistor **410** via a voltage divider circuit **420**. In other words, the gate terminal of the first power transistor **408** is not directly connected to the source terminal of the second power transistor **410**. Rather, the voltage divider circuit **420** indirectly connects the gate terminal of the first power transistor **408** to the source terminal of the second power transistor **410**. The gate terminal of the first power transistor **408** is connected to the voltage divider circuit **420**. The source terminal of the second power transistor **410** is connected to the voltage divider circuit **420**.

[0046] As shown in FIG. 4A, the voltage divider circuit **420** includes a plurality of diodes **412** (e.g., a sequence or a chain of diodes **412**), a capacitor **414-1**, and a resistor **414-2**. The diodes **412** may be an example of the first circuit element **112** of FIG. 1A. In some examples, the voltage divider circuit **420** includes two or more diodes **412** connected in series with each other. In some examples, the diodes **412** includes p-n diodes. In some examples, the diodes **412** include Schottky diodes. The diodes **412** may include a diode **412-1**, a diode **412-2**, a diode **412-3**, and a diode **412-4**. However, it is noted that the number of diodes **412** may be two, three, or any number greater than four. In some examples, the diodes **412** may be configured such as

$$[00002] n \times V_f < V_{gs_max_transistor_first},$$

where n is the number of diodes **412** in the chain. In some examples, the clamping voltage is defined by the sum of the turn-on voltages of the diodes **412**. The capacitor **414-1** and the resistor **414-2** may be an example of the second circuit element **114** of FIG. 1A. In some examples, the resistor **414-2** is connected in parallel with the capacitor **414-1**. In some examples, the combination of the resistor **414-2** and the capacitor **414-1** may provide better voltage sharing across devices at lower frequencies.

[0047] In some examples, the diodes **412** are integrated into the upper device. For example, the semiconductor power device **400** includes a semiconductor package **402** with a first semiconductor device **404** and a second semiconductor device **406**. In some examples, the second semiconductor device **406** is a device that is separate and/or distinct from the first semiconductor device **404**. The second semiconductor device **406** may be connected to the first semiconductor device **404**. In some examples, the first semiconductor device **404** is referred to as an upper device. In some examples, the first semiconductor device **404** is a semiconductor die (e.g., a portion of semiconductor material wafer) that includes the first power transistor **408**. In some examples, the first semiconductor device **404** includes the diodes **412**. In some examples, the capacitor **414-1** and the resistor **414-2** are integrated into the lower device. In some examples, the second semiconductor device **406** is referred to as a lower device. In some examples, the second semiconductor device **406** is a semiconductor die (e.g., a silicon wafer) that includes the second power transistor **410**. In some examples, the second semiconductor device **406** includes the capacitor **414-1** and the resistor **414-2**. In some examples, the diodes **412**, the capacitor **414-1**, and/or the resistor **414-2** can be integrated into one of the power devices (e.g., a semiconductor device with the first power transistor **408** or a semiconductor device with the second power transistor **410**). In some examples, one or more of the diodes **412**, the capacitor **414-1**, and/or the resistor **414-2** may be integrated into a semiconductor device with the first power transistor **408** and one or more of the diodes **412**, the capacitor **414-1**, and/or the resistor **414-2** may be integrated into a semiconductor device with the second power transistor **410**.

[0048] FIGS. 4B and 4C illustrate example waveforms of the semiconductor power device **400**, including the first power transistor **408** and the second power transistor **410**. FIG. 4B illustrates example waveforms without the resistor **414-2**. FIG. 4C illustrates example waveforms with the resistor **414-2**. As shown in FIGS. 4B and 4C, the first power transistor **408** and the second power transistor **410** may operate below their maximum rated voltages, efficiently sharing total cascode device voltage between each other.

[0049] FIG. 5 depicts a flowchart 500 depicting a method with example operations of a semiconductor power device according to an aspect. Although the flowchart 500 is described with reference to the semiconductor power device 100 of FIGS. 1A and 1B, the flowchart may be applicable to any of the semiconductor power devices discussed herein. Although the flowchart 500 of FIG. 5 illustrates operations in sequential order, it will be appreciated that this is merely an example, and that additional or alternative operations may be included. Further, operations of FIG. 5 and related operations may be executed in a different order than that shown, or in a parallel or overlapping fashion.

[0050] Operation 502 includes activating a first power transistor of a semiconductor power device and a second power transistor of the semiconductor power device in response to a gate-source voltage of the semiconductor power device being equal to or greater than a threshold voltage of the second power transistor.

[0051] Operation 504 includes deactivating the second power transistor in response to the gate-source voltage of the semiconductor power device being less than the threshold voltage of the second power transistor.

[0052] Operation 506 includes reducing, by a voltage divider circuit, gate-source voltage of the first power transistor to be less than a gate-source voltage maximum rating (e.g., gate-source breakdown voltage), the voltage divider circuit indirectly connecting a gate of the first power transistor to a source terminal of the second power transistor.

[0053] Clause 1. A semiconductor power device comprising: a first power transistor configured to generate an output voltage; a second power transistor configured to receive an input voltage, the second power transistor connected to the first power transistor in a cascode configuration; and a voltage divider circuit connected to a gate terminal of the first power transistor and a source terminal of the second power transistor.

[0054] Clause 2. The semiconductor power device of clause 1, wherein the voltage divider circuit includes at least one first circuit element and at least one second circuit element.

[0055] Clause 3. The semiconductor power device of clause 2, wherein the at least one first circuit element includes a diode.

[0056] Clause 4. The semiconductor power device of clause 2, wherein the at least one first circuit element includes two or more diodes connected in series.

[0057] Clause 5. The semiconductor power device of clause 2, wherein the at least one second circuit element includes a capacitor.

[0058] Clause 6. The semiconductor power device of clause 2, wherein the at least one second circuit element includes a capacitor and a resistor.

[0059] Clause 7. The semiconductor power device of clause 2, further comprising: a semiconductor package, including: a first semiconductor device that includes the first power transistor; and a second semiconductor device that includes the second power transistor, wherein the first semiconductor device includes the at least one first circuit element and the second semiconductor device includes the at least one second circuit element.

[0060] Clause 8. The semiconductor power device of clause 1, wherein the first power transistor is a normally-on device, and the second power transistor is a normally-off device.

[0061] Clause 9. A semiconductor power device comprising: a first power transistor configured to generate an output voltage; a second power transistor configured to receive an input voltage, the second power transistor connected to the first power transistor in a cascode configuration; and a voltage divider circuit configured to indirectly couple a gate terminal of the first power transistor to a source terminal of the second power transistor, the voltage divider circuit including at least one first circuit element and at least one second circuit element.

[0062] Clause 10. The semiconductor power device of clause 9, wherein the at least one first circuit element is configured to cause a gate-source voltage of the first power transistor to be less than a gate-source maximum voltage rating.

[0063] Clause 11. The semiconductor power device of clause 9, wherein the at least one first circuit element includes a diode.

[0064] Clause 12. The semiconductor power device of clause 9, wherein the at least one first circuit element includes two or more diodes connected in series.

[0065] Clause 13. The semiconductor power device of clause 9, wherein the at least one second circuit element includes at least one of a capacitor or a resistor.

[0066] Clause 14. The semiconductor power device of clause 13, wherein the resistor is connected in parallel with the capacitor.

[0067] Clause 15. The semiconductor power device of clause 9, further comprising: a semiconductor package, including: a first semiconductor device that includes the first power transistor; and a second semiconductor device that includes the second power transistor, wherein the first semiconductor device includes the at least one first circuit element and the second semiconductor device includes the at least one second circuit element.

[0068] Clause 16. The semiconductor power device of clause 9, further comprising: a semiconductor package, including: a first semiconductor device that includes the first power transistor; and a second semiconductor device that includes the second power transistor, wherein the first semiconductor device includes the at least one first circuit element and the at least one second circuit element.

[0069] Clause 17. The semiconductor power device of clause 9, further comprising: a semiconductor package, including: a first semiconductor device that includes the first power transistor; and a second semiconductor device that includes the second power transistor, wherein the second semiconductor device includes the at least one first circuit element and the at least one second circuit element.

[0070] Clause 18. A method comprising: activating a first power transistor of a semiconductor power device and a second power transistor of the semiconductor power device in response to a gate-source voltage of the semiconductor power device being equal to or greater than a threshold voltage of the second power transistor; deactivating the second power transistor in response to the gate-source voltage of the semiconductor power device being less than the threshold voltage of the second power transistor; and reducing, by a voltage divider circuit, a potential of a gate-source voltage of the first power transistor to be less than a gate-source maximum voltage rating, the voltage divider circuit indirectly connecting a gate of the first power transistor to a source terminal of the second power transistor.

[0071] Clause 19. The method of clause 18, wherein the voltage divider circuit includes at least one first circuit element and at least one second circuit element.

[0072] Clause 20. The method of clause 19, wherein the at least one first circuit element includes one or more diodes, and the at least one second circuit element includes at least one of a capacitor or a resistor.

[0073] It will be understood that, in the foregoing description, when an element is referred to as being connected to, electrically connected to, coupled to, or electrically coupled to another element, it may be directly connected or coupled to the other element, or one or more intervening elements may be present. In contrast, when an element is referred to as being directly connected to or directly coupled to another element, there are no intervening elements. Although the terms directly connected to, or directly coupled to may not be used throughout the detailed description, elements that are shown as being directly connected or directly coupled can be referred to as such. The claims of the application, if any, may be amended to recite exemplary relationships described in the specification or shown in the figures. Implementations of the various techniques described herein may be implemented in (e.g., included in) digital electronic circuitry, or in computer hardware, firmware, software, or in combinations of them. Portions of methods also may be performed by, and an apparatus may be implemented as, special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit).

[0074] Some implementations may be implemented using various semiconductor processing and/or packaging techniques. Some implementations may be implemented using various types of semiconductor processing techniques associated with semiconductor substrates including, but not limited to, for example, Silicon (Si), Gallium Arsenide (GaAs), Gallium Nitride (GaN), Silicon Carbide (SiC) and/or so forth.

[0075] While certain features of the described implementations have been illustrated as described herein, many modifications, substitutions, changes and equivalents will now occur to those skilled in the art. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the scope of the embodiments. It should be understood that they have been presented by way of example only, not limitation, and various changes in form and details may be made. Any portion of the apparatus and/or methods described herein may be combined in any combination, except mutually exclusive combinations. The embodiments described herein can include various combinations and/or sub-combinations of the functions, components and/or features of the different embodiments described.

Claims

1. A semiconductor power device comprising: a first power transistor configured to generate an output voltage; a second power transistor configured to receive an input voltage, the second power transistor connected to the first power transistor in a cascode configuration; and a voltage divider circuit connected to a gate terminal of the first power transistor and a source terminal of the second power transistor.
2. The semiconductor power device of claim 1, wherein the voltage divider circuit includes at least one first circuit element and at least one second circuit element.
3. The semiconductor power device of claim 2, wherein the at least one first circuit element includes a diode.
4. The semiconductor power device of claim 2, wherein the at least one first circuit element includes two or more diodes connected in series.
5. The semiconductor power device of claim 2, wherein the at least one second circuit element includes a capacitor.
6. The semiconductor power device of claim 2, wherein the at least one second circuit element includes a capacitor and a resistor.
7. The semiconductor power device of claim 2, further comprising: a semiconductor package, including: a first semiconductor device that includes the first power transistor; and a second semiconductor device that includes the second power transistor, wherein the first semiconductor device includes the at least one first circuit element and the second semiconductor device includes the at least one second circuit element.
8. The semiconductor power device of claim 1, wherein the first power transistor is a normally-on device, and the second power transistor is a normally-off device.
9. A semiconductor power device comprising: a first power transistor configured to generate an output voltage; a second power transistor configured to receive an input voltage, the second power transistor connected to the first power transistor in a cascode configuration; and a voltage divider circuit configured to indirectly couple a gate terminal of the first power transistor to a source terminal of the second power transistor, the voltage divider circuit including at least one first circuit element and at least one second circuit element.
10. The semiconductor power device of claim 9, wherein the at least one first circuit element is configured to cause a gate-source voltage of the first power transistor to be less than a gate-source maximum voltage rating.
11. The semiconductor power device of claim 9, wherein the at least one first circuit element includes a diode.

- 12.** The semiconductor power device of claim 9, wherein the at least one first circuit element includes two or more diodes connected in series.
- 13.** The semiconductor power device of claim 9, wherein the at least one second circuit element includes at least one of a capacitor or a resistor.
- 14.** The semiconductor power device of claim 13, wherein the resistor is connected in parallel with the capacitor.
- 15.** The semiconductor power device of claim 9, further comprising: a semiconductor package, including: a first semiconductor device that includes the first power transistor; and a second semiconductor device that includes the second power transistor, wherein the first semiconductor device includes the at least one first circuit element and the second semiconductor device includes the at least one second circuit element.
- 16.** The semiconductor power device of claim 9, further comprising: a semiconductor package, including: a first semiconductor device that includes the first power transistor; and a second semiconductor device that includes the second power transistor, wherein the first semiconductor device includes the at least one first circuit element and the at least one second circuit element.
- 17.** The semiconductor power device of claim 9, further comprising: a semiconductor package, including: a first semiconductor device that includes the first power transistor; and a second semiconductor device that includes the second power transistor, wherein the second semiconductor device includes the at least one first circuit element and the at least one second circuit element.
- 18.** A method comprising: activating a first power transistor of a semiconductor power device and a second power transistor of the semiconductor power device in response to a gate-source voltage of the semiconductor power device being equal to or greater than a threshold voltage of the second power transistor; deactivating the second power transistor in response to the gate-source voltage of the semiconductor power device being less than the threshold voltage of the second power transistor; and reducing, by a voltage divider circuit, a potential of a gate-source voltage of the first power transistor to be less than a gate-source maximum voltage rating, the voltage divider circuit indirectly connecting a gate terminal of the first power transistor to a source terminal of the second power transistor.
- 19.** The method of claim 18, wherein the voltage divider circuit includes at least one first circuit element and at least one second circuit element.
- 20.** The method of claim 19, wherein the at least one first circuit element includes one or more diodes, and the at least one second circuit element includes at least one of a capacitor or a resistor.
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